

CASE STUDY

HOSPITAL MANAGEMENT SYSTEM

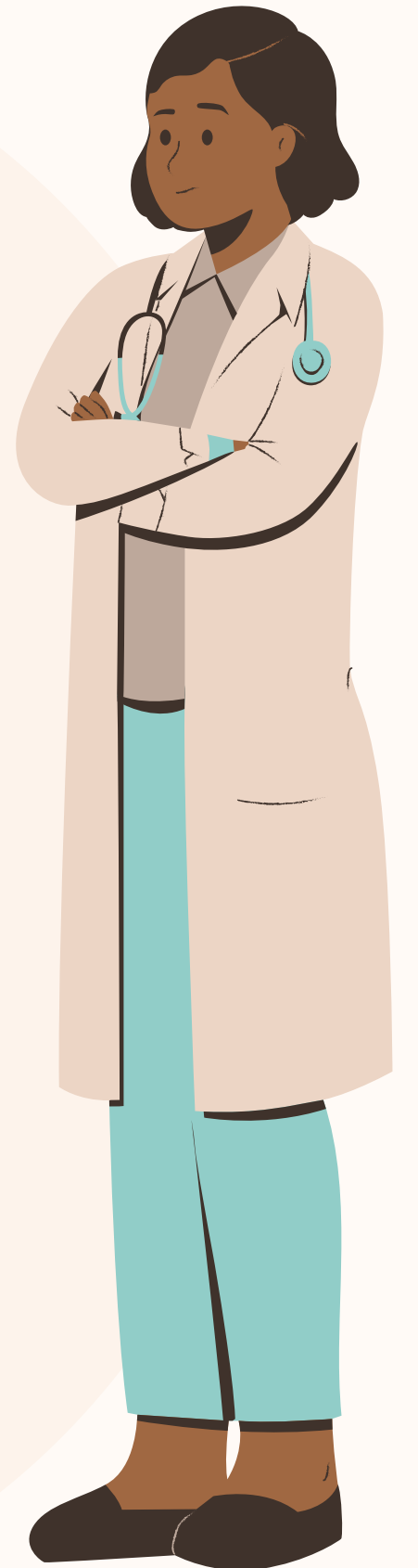


INTRODUCTION

A Hospital Management System (HMS) is a software application designed to manage and organize hospital operations digitally. It helps hospitals and clinics maintain patient records, doctor details, treatments, appointments, and reports in an efficient and secure manner.

Traditionally, hospitals used paper-based record systems, which were time-consuming, difficult to maintain, and prone to errors or data loss. The Hospital Management System solves these problems by storing information digitally and providing quick access to hospital records whenever required.

This project is developed using Python and focuses on simplifying hospital management tasks such as adding patient details, searching records, viewing patient information, and generating reports. The system improves accuracy, reduces paperwork, saves time, and enhances the overall efficiency of hospital operations. The Hospital Management System is useful for hospitals, clinics, and healthcare centers to ensure better patient care and organized data management.



Objectives of Hospital Management System

1) To manage patient records digitally

Store and maintain patient information in an organized digital format.

2) To reduce manual paperwork

Minimize the use of paper records and manual data handling in hospitals.

3) To improve data organization

Keep doctor, patient, treatment, and appointment details properly structured.

4) To provide quick access to records

Enable fast searching, viewing, and updating of hospital information.

5) To increase efficiency in hospital operations

Save time for hospital staff and improve workflow management.

OBJECTIVES

For Better Life



IMPORTANCE OF HMS

Overview 01

- # Reduces paperwork by storing records digitally.
- # Saves time in managing patient and doctor information.
- # Provides quick access to patient records and reports.

Overview 02

- # Improves accuracy and reduces human errors.
- # Prevents data loss caused by damaged or misplaced paper files.
- # Helps manage appointments and treatments efficiently.

Overview 03

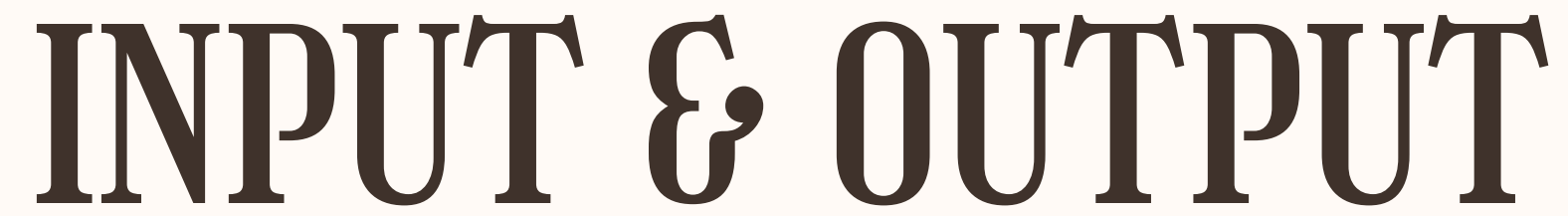
- # Laboratory Management
- # Data Security and Confidentiality
- # Reporting and Decision Making

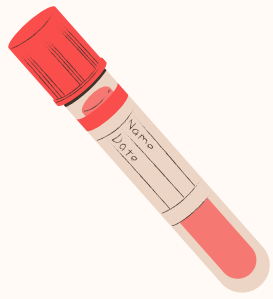
WORKING OF HMS



CODE

```
1  # Hospital Management System in Python
2  patients = []
3
4  def menu():
5      print("\n===== HOSPITAL MANAGEMENT SYSTEM =====")
6      print("1. Add Patient")
7      print("2. View Patients")
8      print("3. Search Patient")
9      print("4. Remove Patient")
10     print("5. Exit")
11
12     while True:
13         menu()
14
15         choice = input("Enter your choice: ")
16
17         # Add Patient
18         if choice == "1":
19             name = input("Enter patient name: ")
20             age = input("Enter patient age: ")
21             disease = input("Enter disease: ")
22
23             patient = {
24                 "Name": name,
25                 "Age": age,
26                 "Disease": disease
27             }
28
29             patients.append(patient)
30             print("Patient added successfully!")
31
32         # View Patients
33         elif choice == "2":
34             if len(patients) == 0:
35                 print("No patient records found.")
36             else:
37                 print("\nPatient Records:")
38                 for i, patient in enumerate(patients, start=1):
39                     print(f"\nPatient {i}")
40                     print("Name      :", patient["Name"])
41                     print("Age       :", patient["Age"])
42                     print("Disease  :", patient["Disease"])
43
44         # Search Patient
45         elif choice == "3":
46             search_name = input("Enter patient name to search: ")
47
48             found = False
49
50             for patient in patients:
51                 if patient["Name"].lower() == search_name.lower():
52                     print("\nPatient Found")
53                     print("Name      :", patient["Name"])
54                     print("Age       :", patient["Age"])
55                     print("Disease  :", patient["Disease"])
56                     found = True
57                     break
58
59             if not found:
60                 print("Patient not found.")
61
62         # Remove Patient
63         elif choice == "4":
64             remove_name = input("Enter patient name to remove: ")
65
66             found = False
67
68             for patient in patients:
69                 if patient["Name"].lower() == remove_name.lower():
70                     patients.remove(patient)
71                     print("Patient removed successfully!")
72                     found = True
73                     break
74
75             if not found:
76                 print("Patient not found.")
77
78         # Exit
79         elif choice == "5":
80             print("Exiting Hospital Management System...")
81             break
82
83         else:
84             print("Invalid choice. Please try again.")
```

A stylized, flat-design illustration of a microscope. The main body is a light beige color. The eyepiece is at the top, with a dark brown barrel. Below it is the objective lens assembly, featuring a large, dark brown circular lens and a smaller, lighter brown circular lens. The stage is a horizontal platform with a dark brown base and a lighter brown top. The slide holder is a dark brown rectangular piece. The base of the microscope is a large, light beige curved shape. The entire illustration is set against a white background.



CONCLUSION

The Hospital Management System (HMS) is an effective software solution that helps hospitals and clinics manage their daily operations efficiently. It simplifies tasks such as patient registration, appointment scheduling, record maintenance, billing, and report generation. By automating these activities, HMS reduces paperwork, saves time, minimizes errors, and improves the quality of healthcare services.

An HMS also enhances communication between different departments like laboratories, pharmacies, and administration, leading to better coordination and faster patient care. Overall, the system improves hospital productivity, ensures secure data management, and provides better satisfaction for both patients and hospital staff.

